

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	↑						↓
Frequency (ν)	↑						↓
Wavelength (λ)	↓						↑
Wavelength (λ)	<0.01	0.01-10	10-400	400-700	700-1e6	1e6-9	>1e9

• violet • indigo • blue • green • yellow • orange • red

(2) $c = \lambda\nu$ (Speed of light = wavelength × frequency)

<i>c</i>	ν (Hz)	kHz	MHz	GHz
λ (m)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck × frequency)

hc , <i>h</i>	λ (m)	λ (nm)	ν (Hz)
E (J / photon)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

• Grams → mol → ×6.022e23 for photon **count**

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electrons are quantum (line spectra); Rydberg eq.
De Broglie	Electrons are particle and wave (5)
Schrodinger	Wave equation (energy and spatial prob. of e-) (n, l, m_l, m_s)
Heisenberg	Position and velocity not both knowable

• $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$

• $n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}$; $n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$

Variable	Value	Variable	Value
<i>h</i>	6.626e-34 J·s	<i>h/R</i>	3.0394e-16 s
<i>c</i>	2.9979e8 m/s	<i>hc</i>	1.9864e-25 J·m
<i>R</i>	2.18e-18 J	<i>hc/R</i>	9.1120e-8 m

(5) $\lambda = h/(m \cdot v)$ (De Broglie)

- 1 nm = 1e-9 m
- 1 e- = 9.11e-31 kg

<i>h</i>	u (m/s)	km/s
m (kg)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

(6) Electron Configuration (Aufbau, Pauli, Hund)

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p
- Round-robin sublevel before double occupaton (Hund)
- Two e- per orbital (Pauli's exclusion)

Symbol	Name	Value
<i>n</i>	Principal energy level (shell)	(1, 2, 3, ...)
<i>l</i>	Sublevel = subshell (shape)	(0, ..., n - 1)
<i>m_l</i>	Orbital (orientations)	(-l, ..., l)
<i>m_s</i>	Spin (2 per orbital)	±1/2

Sublevel	<i>l</i>	<i>m_l</i>	# orbitals	# e-
s	0	0	1	2
p	1	±1	3	6
d	2	±2	5	10
f	3	±3	7	14

Element	EC	Element	EC
Cr	[Ar]4s ¹ 3d ⁵	Ag	[Kr]5s ¹ 4d ¹⁰
Mo	[Kr]5s ¹ 4d ⁵	Au	[Xe]6s ¹ 4f ¹⁴ 5d ¹⁰
Cu	[Ar]4s ¹ 3d ¹⁰	Pd	[Kr]4d ¹⁰ (no 5s)

(7)–(8) Reading EC, Anion, Cation

EC → element	add every superscript to get <i>Z</i>
Paramagnetic	at least 1 unpaired val. e- (else dia.)
Ground state	follows Aufbau (else excited)
Valence electrons	highest / outer shell (else core)
Anion	count val. e- needed until noble gas
Cation	remove val. e- until noble gas
Tx metal	remove s before d
Isoelectric	same EC (e.g. Na ⁺ , Al ³⁺ , F ⁻ , N ³⁻ , Ne)

(9) Quantum Numbers, Penetration & Shielding

- Shell *n* holds *n*² orbitals, 2*n*² e-
- **energy: s < p < d < f** (within shell, same *n*)
- **Pauli exclusion** — no two e- carry the same four quantum numbers → 2 per orbital, opposite spins
- **Penetration** — orbital reaching into the core region
- ↑ shielding ⇔ ↑ reduced *Z_{eff}* ⇔ ↓ penetration ⇔ ↑ **energy** (reverse also holds)
- **Shielding** — only **core** (inner) e- shield effectively; valence e- shield each other poorly

(10) *Z_{eff}* & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e- = total e- – valence e- (= the **previous noble gas**)
2. $Z_{eff} \approx Z - \text{core } e^-$
 - Same period → same core, so the **rightmost** element wins
 - Poor valence shielding is why *Z_{eff}* rises across a period

(11)–(13) Trends

	Trend	Top	Bot	Left	Right
(11) Atomic size (radius)	↙	↓	↑	↑	↓
(12) Ionization energy	↗	↑	↓	↓	↑
(13) Electron affinity	↗	↑	↓	↓	↑
<i>Z_{eff}</i>	↘	↓	↑	↓	↑
Metallic	↙	↓	↑	↑	↓
Nonmetallic	↗	↑	↓	↓	↑
Electronegativity (EN)	↗	↑	↓	↓	↑

Atomic size

- Down a group **bigger** (new shell) · across a period **smaller** (*Z_{eff}* pulls in)
- Group position beats period position when both apply
- cation < neutral < anion

Ionization energy

- He ↑ of all · opposite of radius
- *IE₂* > *IE₁* always
- **Big jump** appears once the first **core** e- is removed

Electron affinity

- **Most eager for an e- = halogens, upper right** — this course writes it both as "most negative *EA*" and "largest positive *EA*"; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- ***EA* (X) and *IE* of X are the same magnitude** — F + e- releases exactly what F → F costs

(14) Families & Diagonal Rule

Family	Name	Reactivity	e-	Charge
1A	alkali metals	most reactive metals (downward)	lose 1	1+
2A	alkaline earth	reactive, less than 1A	lose 2	2+
7A	halogens	reactive nonmetals (upward)	gain 1	1-
8A	noble gases	inert	—	0

- **Lanthanide** (58–71), **Actinide** (90–103)
- **Diagonal rule** (e.g. $\text{Li} \sim \text{Mg}$, $\text{Be} \sim \text{Al}$, $\text{B} \sim \text{Si}$)
- **Diatomic** (H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 (l), I_2 (s))

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K_2O , MgO , Fe_2O_3 , Cr_2O_3) · nonmetal oxide = **acidic** (P_4O_{10} , SO_2 , CO_2 , Cl_2O_7) · Al_2O_3 and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3- · 6A 2- · 7A 1-; criss-cross and reduce

• Same-group swap keeps the pattern: $\text{BaS} \rightarrow \text{CaO}$

• $2\text{A} + 3\text{F}_2 \rightarrow 2\text{AF}_3 \Rightarrow \text{A is 3+ (Al)} \cdot 3\text{Li} + \text{Z} \rightarrow \text{Li}_3\text{Z} \Rightarrow \text{Z is 3-}$, so Mg gives **Mg₃Z₂**

Bond-type ID — ionic and covalent together = a salt of a **polyatomic ion** (NH_4NO_3 , K_2SO_4); all-nonmetal = covalent only

Family reactions

- $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$
- $\text{H}_2\text{(g)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{HCl(g)}$

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first

2. If charges tie, smaller ions win

- $\uparrow$ charge $\rightarrow \uparrow$ **U** (dominant) · $\uparrow$ ionic radius $\rightarrow \downarrow$ U

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read EN off the chart — $EN \uparrow$ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)

2. Classify

~ 0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar"** $\rightarrow \Delta EN$ **alone** (largest difference wins)
- **"Strongest / shortest bond"** $\rightarrow$ **bond order FIRST**: triple > double > single. ΔEN only breaks ties between equal orders — a $\text{C} \equiv \text{N}$ beats a high- ΔEN $\text{Si}=\text{O}$
- Larger $\Delta EN \rightarrow$ stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. FC on every atom

2. Check ΣFC = the overall charge

3. Pick the structure with FC 's **nearest 0**

4. Put any -1 on the **more electronegative** atom

- Equivalent resonance forms $\rightarrow$ real bonds are the **average** (e.g. $1\frac{1}{2}$)

(19) Lewis Structures x4

1. Total valence e- — +1 per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e- onto the central atom
6. Central below 8 $\rightarrow$ pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e- per atom (H wants 2) — H_2O , CH_4 , NH_3 , CO_2 , CH_2Cl_2 , $\text{CH}_3\text{CH}_2\text{Cl}$
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. $\text{BF}_3/\text{BCl}_3/\text{AlI}_3$ leave 6 e- on the center; BeCl_2 leaves 4
(c) Expanded octet	Central atom period 3 or below (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — AsCl_5 , XeI_4 , SO_4^{2-}
(d) Organic	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH₂ amine

- Every C makes 4 bonds; a -COOH carbon carries one =O and one -O-H; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero FC , add double bonds to zero it (e.g. two S=O in SO_4^{2-})

(20) Photoelectric Effect ($h\nu$ - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

Variable	Value
h	6.626e-34 J·s
m_e	9.11e-31 kg
hc	1.9864e-25 J·m
$2hc/m_e$	4.3609e5 m ³ /s ²

Have	Want	Use
λ , λ_{thr}	u	$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
ν_{thr} or λ_{thr}	Φ , Φ / mol	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$, n.b. $\Phi \times N_A$ (/1e3 for kJ/mol)
Φ , λ	u	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)

2. $\Delta x \geq h/(4\pi \cdot \Delta p)$

- Result is the **minimum** uncertainty
- Rearrange the same way to solve for Δu
- Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), IE (+), bond dissociation (+), EA (usually -), U (-)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements $\rightarrow$ gaseous atoms $\rightarrow$ gaseous ions $\rightarrow$ solid
4. Set the chain equal to ΔH°_f
5. Solve for the single unknown
 - Use IE_2 as well when the cation is 2+
 - U runs ions(g) $\rightarrow$ solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. $\Sigma BE(\text{broken})$, always positive
3. $\Sigma BE(\text{formed})$, subtracted
4. $\Delta H = \Sigma \text{broken} - \Sigma \text{formed}$
 - Bonds unchanged on both sides cancel — skip them
 - Negative ΔH = exothermic
 - For "heat released by g of X," convert g $\rightarrow$ mol and scale